

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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19 September 2026

Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements and state the conditions that connect them: *coherence*, meaning collective phase order; *compatibility*, meaning that local descriptions agree where their domains overlap; and *self-representation*, meaning that the resulting state carries an accurate reconstruction of itself. Four results carry the argument. Installed energy bounds the mean coupling a declared actuator supplies, so a coherent stationary branch requires it to exceed $2D/\kappa$, with D the phase diffusion and κ set by the mode prices. Phase encoders shared on overlaps turn phase order into a bound on content disagreement, vacuous for a whole population and informative within a patch. Compatible local states glue to a unique global state. A contracting reconstruction that fixes that state makes it the unique fixed point, while accuracy across a family of states requires distinguishable encodings. Lean 4 checks these results and their composition, with the eight connecting hypotheses as explicit arguments and no axioms beyond its libraries'. Numerical controls bound what observation can decide: distinct mechanisms share stationary phase

distributions, finite ramps shift the apparent onset exponent, a winding field and a uniform one are stationary at one coupling with resultants spanning nearly the range of that observable, and a coupling rule descending a squared-drift objective holds coherence while losing the structure it was meant to acquire. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling. Testing it requires calibrated geometry, resolved phase dynamics, validated decoders and a spatial observable beside the resultant.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Unity has been approached differently elsewhere: integrated information

17 theory treats integration and exclusion as constitutive (Tononi et al., 2016),
18 global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene
19 et al., 2014), and predictive-processing accounts derive both from a hierar-
20 chical generative model (Friston, 2010). Here the three requirements are
21 kept apart rather than derived from one another, so that each carries its own
22 assumptions and admits its own measurement; Section 8.4 returns to the
23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand’s relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions
30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these
33 requirements and the assumptions each connection needs. A declared ac-
34 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
35 overlaps convert phase order into a bound on content disagreement. Compat-
36 ible spatial measures have a unique global extension; approximate agreement
37 controls a specified global selection. A contracting reconstruction makes that
38 extension its unique fixed point only if it accurately reconstructs the exten-
39 sion. Accuracy across a family of states imposes a further condition on the

40 information retained by the encoding.

41 The empirical contribution is to identify what observations can decide.
42 Stationary phase distributions can be shared by stronger coupling, weaker
43 diffusion and a matched common drive. Finite ramps change the observable
44 onset. A winding field is stationary at the same coupling as a uniform one and
45 has the opposite resultant, so that observable does not determine the state.
46 Coherent adaptation can lose alignment with a designated environmental
47 template, and a compatibility estimator needs diagnostics that distinguish
48 content from decoder artifacts. These findings determine the measurements
49 needed to test the proposed connections.

50 Three of the ingredients are standard: the synchronization threshold
51 $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing
52 of compatible local sections, and the Banach fixed-point theorem (Banach,
53 1922). Four things here are not. The installed-energy condition bounds
54 mean coupling by the stored energy of a declared actuator, which makes a
55 resource condition necessary for coherence on the stationary branch. The
56 coherence-to-content bound converts phase order into a statement about de-
57 scriptions rather than about timing, and its patch-local form is what keeps
58 it from being vacuous in a large population. A stability criterion on the
59 reweighted coupling kernel, together with an existence condition on an am-
60 plitude layer, settles when a low resultant reports weak coupling and when
61 it reports a winding field. And the composition is machine-checked together
62 with a witness discharging all eight hypotheses at once, which is what makes

63 their conjunction non-empty rather than a list no object satisfies.

64 Granting those premises, collected in Table 1, the account runs as follows.
65 Extracellular contraction on awakening raises endogenous field amplitude
66 and with it the effective coupling between cortical oscillators. Once that
67 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
68 within a quarter turn the population settles onto it; other initial data leave
69 windings stationary at the same coupling. Encoders shared where cortical
70 territories overlap turn that phase order into agreement about the quantities
71 the territories describe in common, so the local descriptions glue: there is
72 one state of which each is a restriction. Part of that state encodes the state
73 itself, and a readout of that part reconstructs it accurately enough, and
74 sensitively enough across the states that matter, to make it the unique state
75 the system models itself as being. On this account the glued state is what
76 makes distributed activity one situation, and its accurate self-reconstruction
77 is what gives that situation a perspective. Every step of it is conditional.

78 Sections 2–4 develop the three requirements together (Fig. 1). Section 5
79 composes them, stating the thermodynamic constraints and all eight con-
80 necting hypotheses per branch; its supporting models of feedback, learning
81 and resource supply have distinct state spaces and assumptions. Section 6
82 then examines what the observables can establish, before Section 7 devel-
83 ops the cortical hypothesis and an awakening experiment. The hypothesis is
84 that extracellular contraction increases field-mediated coupling; its direction,
85 rate and relation to recovery require independent calibration. Supplemen-

86 tal Material organizes the constructions and controls behind the claims they
 87 support.

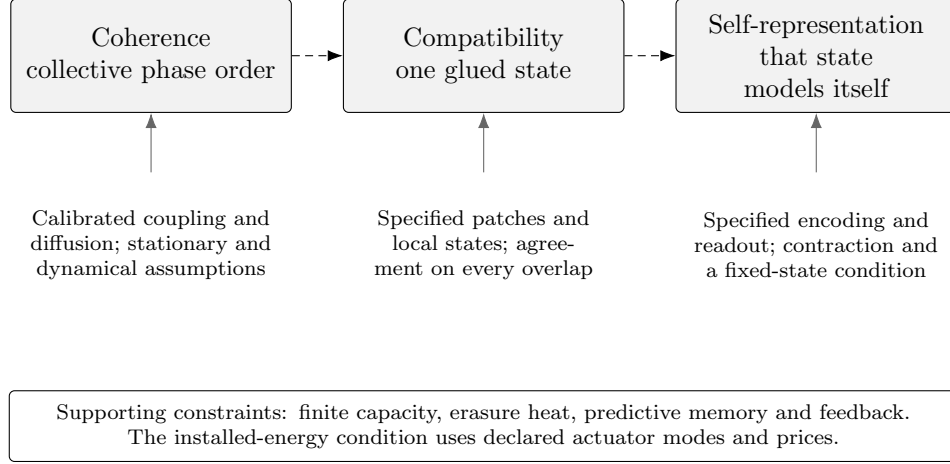


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

88 2 Coherence: physical coupling and its re- 89 source condition

90 2.1 Dynamics and the stationary threshold

91 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

92 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
93 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
94 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
95 parameters. The stationary calculation below uses identical frequencies and
96 this scalar coupling. Applying it to a spatial kernel requires an aggregation
97 rule and a check of the resulting approximation.

98 For identical frequencies in their rotating frame, the positive normalized
99 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

100 where I_n are modified Bessel functions and ψ is the collective phase. The for-
101 mal derivation uses the absence of stationary current for a periodic gradient
102 drift to obtain this density, then verifies normalization and self-consistency.
103 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above

threshold, exactly one positive solution coexists with zero and increases
strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form is strictly dissipative transverse to collective rotation, whose rate is zero. These linear estimates concern the limiting equation. Relating them to finite stochastic trajectories requires a dynamical mean-field limit and branch selection; exchange symmetry alone does not establish the needed independence (Sznitman, 1991).

2.2 Installed energy bounds mean coupling

A spatial model replaces the finite sum by integration against a kernel. One declared implementation carries response modes with continuous profiles ϕ_i , installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

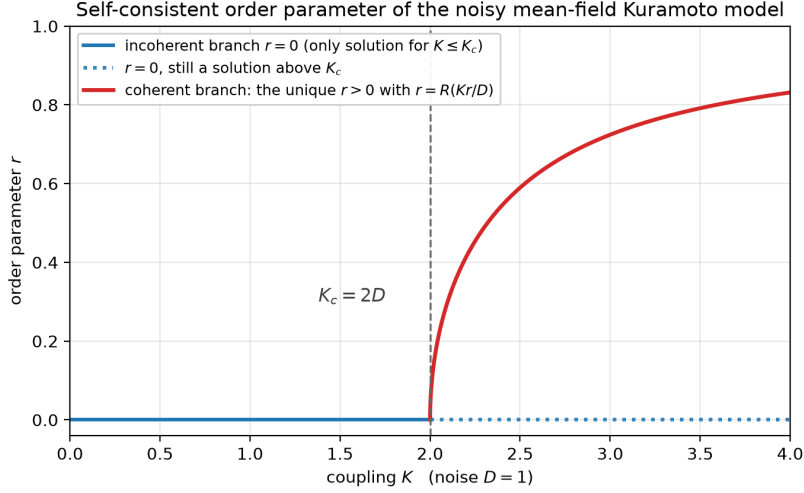


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

119 The same configuration sets the reciprocal kernel and stored energy. An
 120 executed transition changes the kernel through its changed occupancies and
 121 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 122 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 123 pair reconstruction on refining spatial partitions converges to its product-
 124 measure integral under the specified compactness and sampling assumptions;
 125 the refinement index is independent of the number of control updates.

126 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 127 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

128 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_z p(z) U(z), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned}
 \tag{5}$$

129 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 130 coherent stationary branch and any growing Fourier mode about the uniform
 131 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 132 is necessary: energy above it need not produce sufficient coupling because
 133 the bound can be loose.

134 The quantity constrained is installed energy. Dissipated heat does not
 135 determine it, and κ is supplied by the mode profiles, prices and substrate
 136 normalization. Applying the scalar threshold to heterogeneous or signed spa-
 137 tial dynamics requires a justified reduction. Identifying the installed kernel
 138 with a cortical field is a separate physical commitment.

139 **3 Compatibility: from coordinated activity** 140 **to a global description**

141 **3.1 Why common timing is insufficient**

142 For overlapping patches U and V , compatibility means that restricting either
 143 local description to $U \cap V$ gives the same object. Descriptions can agree on

144 shared quantities while contributing different information elsewhere. Com-
 145 patible sections of the specified sheaf have a unique global extension. On
 146 a finite cover, local finite spatial measures agreeing as measures on every
 147 overlap also have a unique finite-measure extension whose sheaf image is the
 148 glued section. No mass-one normalization is imposed.

149 In the cup example, visual and reaching descriptions must express a
 150 shared quantity, such as cup location, in common coordinates. Restriction
 151 retains that quantity while dropping details particular to each description.
 152 Agreement permits surface appearance and hand configuration to contribute
 153 to one situation. The mathematical counterexample assigns the same phase
 154 to every patch while two local mass profiles disagree on a shared site. Those
 155 profiles have no common extension. Because they are chosen independently
 156 of phase, synchrony places no constraint on their disagreement.

157 This distinction limits the biological interpretation. Receptive territories
 158 and content domains must be related to spatial patches, and the descriptions
 159 must support the required extension. Matching overlap marginals of de-
 160 coded probability laws need not determine a unique global joint distribution
 161 (Abramsky & Brandenburger, 2011).

162 **3.2 When coherence does constrain content**

163 A declared relation between phase and content gives a positive result. Let
 164 contents take values in a real normed vector space E . Patch i reads an
 165 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point

166 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 167 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 168 sites; their equality on overlaps is a supplied relation between descriptions of
 169 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 170 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 171 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

172 Here discrepancy is the norm of the difference. At $r = 1$, the observa-
 173 tions agree exactly. The factor N makes this estimate potentially weak
 174 in large populations; it supplies a pointwise guarantee from a population
 175 average. The population in it need not be the whole array. A patch car-
 176 ries an order parameter of its own, and the same bound read there car-
 177 ries that patch's resultant and that patch's site count (`chord_le_of_patch_`
 178 `coherence, compatible_of_patch_coherence`), so a large population ob-
 179 served through small patches is constrained by the small count. A shared
 180 scalar encoder is a special case. Dropping the overlap relation permits dis-
 181 agreement even at perfect locking. A common nonexpansive linear projection
 182 preserves agreement, but agreement of a projection does not establish agree-
 183 ment of the full vector.

184 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 185 ηo_i . Content and observation discrepancies bounded by ε and δ give a next

186 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 187 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

188 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 189 duce an initial mismatch. Approximate observations leave a nondecaying
 190 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 191 update to the phase process.

192 **3.3 What approximate agreement determines**

193 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 194 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 195 Weights that vanish outside their patches and sum to one at each site select
 196 a global profile by weighted averaging. The selection lies within ε of each
 197 patch there, and any two such selections differ by at most ε , independently
 198 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 199 $L\varepsilon$ across selections.

200 At zero tolerance, the selection is the unique exact extension. At every
 201 positive tolerance there are families with no exact extension; selection then
 202 gives an approximation whose dependence on weights is bounded. Supple-
 203 mental Material gives a three-site witness attaining the bound. The mass-
 204 profile construction uses the scalar case of the content dynamics and addi-
 205 tionally requires nonnegativity of the initial profiles and observations. The

206 general vector result supplies no identification with the spatial-measure sheaf.

207 These constructions concern declared objects: a specified cover, contents
208 valued in a normed space, and encoders related on overlaps. They select
209 neither the neural cover nor the decoder. Identifying their contents with
210 perceptual content, and establishing that the identification bears on experi-
211 ence, is a further modelling task; Section 6 sets out what a measurement of
212 it would have to establish.

213 4 Self-representation: a state that models it- 214 self

215 4.1 The reflexive construction

216 Compatibility gives a global state. Reflexivity relates it to an encoding within
217 that state. Equip global sections with a complete metric. If the additional
218 restriction-resonance condition identifies encoding with restriction to a region
219 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

220 The region and readout are fixed across the state space. If F contracts and
221 fixes the section s_* formed from the local descriptions, Banach's theorem
222 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89

223 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
 224 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
 225 one for $\tau > 0$. Constructing a map with that factor remains a modelling
 226 obligation.

227 In the cup example, the global state includes the represented scene and the
 228 perceiver's relation to it. A local encoding must retain enough information
 229 for the readout to recover that modelled state; cup position alone would be
 230 insufficient if hand configuration and viewpoint could not be reconstructed.

231 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
 232 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (9)$$

233 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 234 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 235 weakens.

236 4.2 Reconstruction across a family of states

237 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 238 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 239 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

240 Thus M relevant states pairwise more than 2ε apart require at least M
 241 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 242 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 243 existing self-prediction map.

244 This is an additional accuracy criterion on a family, with no requirement
 245 for multiple exact fixed points. A blind encoder or constant readout fails it
 246 for separated states despite having a unique fixed point.

247 **4.3 Agreement after a change takes communication**

248 Snapshot compatibility does not specify how quickly descriptions respond
 249 when a shared quantity changes. In a synchronous network whose updates
 250 read only the site’s own state and declared neighbours’ messages, two initial
 251 configurations agreeing on a site’s causal past through T rounds produce the
 252 same state there at round T . If two fresh interventions outside that past
 253 require reports separated by more than 2ε , no report from that site can be
 254 within ε of both. This follows from the reconstruction bound in Section 4.2.

255 A three-site delay line witnesses failure after one round and exact report-
 256 ing after two; cutting the path prevents delivery at every round. A report
 257 may still happen to be correct about one fixed world. The bound concerns
 258 responsiveness across interventions.

259 Throughout this section *prediction* names a reconstruction map; reading
 260 it as temporal neural prediction would require dynamics. The relevant family
 261 of states, the metric, the tolerance, the communication graph and the dead-

line are all declared inputs, as are the represented variables and the fidelity counted as accurate. Reading the resulting code count as physical memory capacity requires a hardware model, and a readout that varies with the state requires a further one.

5 Thermodynamics and composition

The phase and representation results concern coupling, content disagreement and reconstruction error. Thermodynamics constrains the physical operations that could implement them. Finite-register capacity obeys $H(p) \leq \log |X|$; erasure heat depends on specified system–bath assumptions (Landauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result concerns a field reaching disconnected vacuum components (Kibble, 1976). The composition connects these objects through the explicit hypotheses in Table 1.

5.1 Feedback changes the information constraint

For passive memory X_t driven by an environmental signal S_t , the predictive bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still et al., 2012). Zero retained information also satisfies it, so a task and a

280 learning rule require additional specification. Relating this bound to Lan-
 281 dauer heat requires a common thermal scale and budget, and memory waste
 282 bounded by erased entropy.

283 An acting agent changes the signal it will receive. Specify policy, world,
 284 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned}
 \tag{12}$$

285 The second conditional term vanishes in the passive model. With feedback,
 286 the signed difference can be negative as action creates correlations with the
 287 future. A causal reading requires an environmental state sufficient to describe
 288 the relevant dynamics.

289 For a controlled substep, the active thermodynamic bound instead uses
 290 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 291 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
 292 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
 293 2013). Every forward path of positive probability must have a positive-
 294 probability reverse for this finite identity. A missing reverse gives infinite
 295 divergence. The supplement develops the support conditions and composes
 296 actuation with memory updates on their shared intermediate law.

297 5.2 Learning and repeated operation need explicit re- 298 sources

299 The finite examples distinguish selecting a feasible policy, relaxing a register
300 toward an encoded reward, and learning an unknown task from observa-
301 tions. The observational learner acquires correlation with an environmental
302 parameter while its action marginal stays uniform. A continuing model exe-
303 cutes actuation, observation-driven learning and reset in sequence, preserving
304 above-chance performance at a positive work cost per cycle.

305 A phase-derived sensor makes the informational distinction concrete: an
306 informative readout and a blind control have the same phase order but dif-
307 ferent learning performance. Attaching the learner’s register to the actuator
308 of Section 2.2 ties reward, installed kernel and work to its own executed law.
309 The learning witness rises above the necessary installed-energy bound while
310 its matched control remains on the exclusion line. These finite constructions,
311 detailed in Supplemental Material, leave the evolving feedback between phase
312 dynamics, installation and sensing to be specified.

313 For that model, common energy E , thermal scale θ , local detailed balance
314 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

315 where expected cumulative work stays within the supplied allowance B at
316 every prefix. This bounds expectations. Separate constructions track a store

on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles, while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 2.2 requires physical relations that these separate models do not supply automatically.

5.3 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.

Table 1 (continued)

Connection	Status	Required assumption
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 2.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are independent roots rather than a step from one to the other; both E45 discharges take their convergence from refinement data and not from the thermodynamic bound; E78 states that coherence relaxes the supplied cover, and the coherence-to-content estimate of Eq. (6) bounds disagreement only for encoders already shared on overlaps. The inferential content of the composition is its last step, where contraction and a fixed glued section give uniqueness. What composing them establishes is that the premises are consistent, named

350 in one place and machine-checked together, and not that coherence follows
351 from installed energy or self-representation from coherence.

352 Joint witnesses combine a register, boundary, refining mesh and small
353 spatial cover; active witnesses add controlled thermal updates and a priced
354 installation. They establish satisfiability under their stated identifications.
355 Mesh convergence follows from refinement data, and the witnesses establish
356 no common cortical mechanism. Supplemental Material maps the claims to
357 their Lean declarations and records the counterexamples. The development
358 declares no additional axioms beyond its libraries' foundational axioms.

359 **6 What the observations establish**

360 The composition holds under its premises. Whether observation can establish
361 those premises is a separate question, and the three requirements it connects
362 concern different objects: phase dynamics, local descriptions and internal
363 encodings. Their observables must retain these distinctions. The following
364 controls show which mechanisms remain indistinguishable, which temporal
365 approximations fail, which states one observable cannot separate, and what
366 independent content measurements must establish.

367 6.1 Stationary distributions leave the mechanism un- 368 determined

369 Equation (2) gives a concentration–coherence relation with no fitted shape
370 parameter. Estimate r from circular resultant length and a from density
371 shape: inverting R from the measured r would make agreement automatic.
372 Even independently estimated agreement identifies a statistical family shared
373 by different mechanisms.

374 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
375 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
376 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
377 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
378 and diffusion D_0 have that same density. Their stationary distributions co-
379 incide window by window despite different causes.

380 Resolved dynamics supply additional information. In the ideal model with
381 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
382 increment means constrain drift. Separating endogenous coupling from a
383 matched common drive additionally requires a measured drive or controlled
384 perturbation; identifying electromagnetic coupling requires evidence specific
385 to that interaction. Supplemental Material gives the equivalence construction
386 and its numerical check.

387 6.2 Finite ramps change the observable onset

388 The square-root law in Eq. (3) describes stationary solutions. A changing
389 coupling requires time for the phase distribution to approach those solutions.
390 In a deterministic mean-field control that crosses the threshold symmetrically,
391 none of the scanned rates meets the declared tracking tolerance: slowing
392 the crossing does not remove the terminal residual under that initialization.
393 A slow control variable therefore supplies no general guarantee of branch
394 tracking at onset. Supplemental Material gives the residual, rate range and
395 admissible rates away from threshold.

396 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
397 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
398 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
399 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
400 6 uncensored legs supply the delay scaling below; the decade-spaced ones
401 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
402 onset exponents differ from those of the exact stationary branch evaluated
403 on the same grids. At the fastest speed the ensemble mean never reaches the
404 fit window, so its fitted slope does not estimate stationary onset.

405 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
406 makes the apparent exponent depend on ramp speed and fitting window.
407 The specified deviation-based critical ramp speed is not bracketed by this
408 sweep. With fluctuating coupling, the relevant stationary comparison also
409 depends on rate: the mean coupling approximates order in the tested fast

410 regime, while averaging stationary order over coupling works for the tested
 411 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 412 neither approximation. These are finite-horizon observations, with controls
 413 and parameters in Supplemental Material.

414 The delay scaling is the prediction this section supports, and it needs
 415 no absolute calibration. Over the 6 uncensored legs the mean delay follows
 416 $\Delta K \propto v^{0.444}$, with a 95% slope interval of $[0.394, 0.494]$ that excludes the
 417 predicted $1/2$. The deterministic mean-field control places that shortfall
 418 in the delay estimator rather than in finite-population sampling. Stated
 419 as a law rather than as a number: onset delay scales as the square root
 420 of the rate at which coupling crosses threshold. Testing it requires only
 421 that $\dot{K}$ be monotonic in a manipulated rate — not its value, nor κ , nor γ
 422 — so it is a within-subject design that varies the rate of emergence from
 423 anaesthesia alone, with each participant as their own control. Such a design
 424 must avoid the hazard this sweep exhibits: a fixed absolute escape criterion
 425 meets a size-dependent critical fluctuation floor, so the criterion has to be
 426 referred to each recording’s own pre-threshold fluctuation scale rather than
 427 to a shared number. The diffusion D is the remaining input, and Section 6.1
 428 notes that it is estimable from resolved phase traces through the rate $2D$
 429 at which quadratic variation accumulates, which is a different quantity from
 430 the concentration the propofol exercise could not range over.

431 Additional controls address integration bias, population size and spatial
 432 or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in$

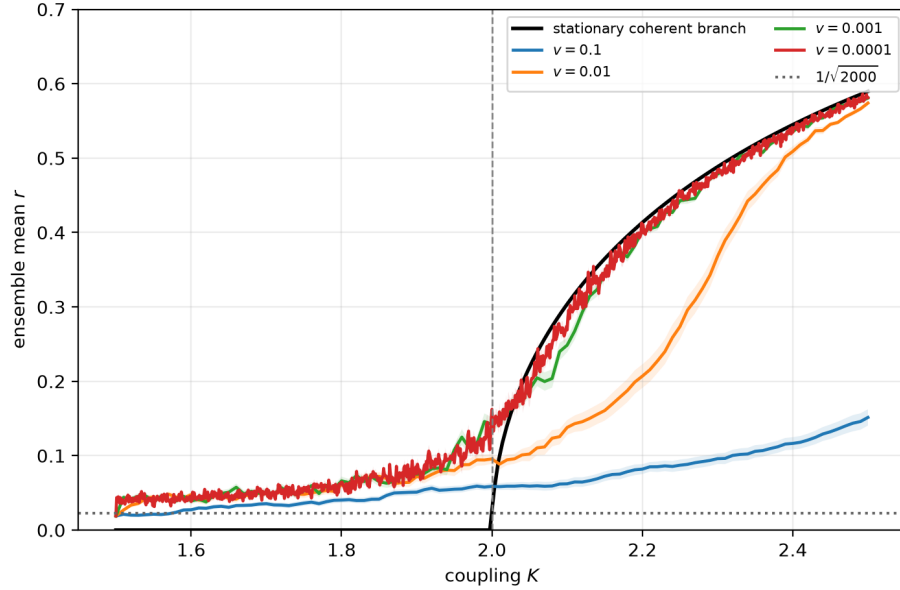


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

433 [1.958, 2.009] contains 2. These are finite numerical regimes, with full pa-
 434 rameters and controls in Supplemental Material; they supply no cortical cal-
 435 ibration.

436 **6.3 A winding state leaves the order parameter am-** 437 **biguous**

438 The order parameter r is indifferent to where a phase sits in space. A field
 439 winding once across a periodic sheet therefore has $r = 0$ while every patch

440 of it stays locked. Such a field is not an artificial construction here. On a
 441 torus a winding is stationary for any isotropic symmetric kernel at identical
 442 frequencies: the coupling term is odd in the separation against a kernel even
 443 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
 444 this for any finite periodic array, this sheet included. A winding satisfies the
 445 positivity and identical-frequency hypotheses. It violates only the quarter-
 446 turn initial spread that the convergence result additionally requires before
 447 placing the limit at the uniform state.

448 On the sheet of Section 2's spatial control, a winding survives below
 449 an interpolated decay length of 0.1871 mm and unwinds above it. That
 450 boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the
 451 same coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start
 452 and $r = 0.0083$ from a winding one. Both are stationary, with patch-local
 453 order 0.9015 and no winding defect. At fixed K the observable therefore takes
 454 two values spanning nearly its whole range. Coupling does not determine the
 455 resultant, and a low resultant does not by itself distinguish weak coupling
 456 from a winding field.

457 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-
 458 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 459 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
 460 Heterogeneity shortens the range that sustains a wave without removing it.

461 Which windings persist has a criterion behind it. Linearizing the phase
 462 model about a winding returns a coupling of the same translation-invariant

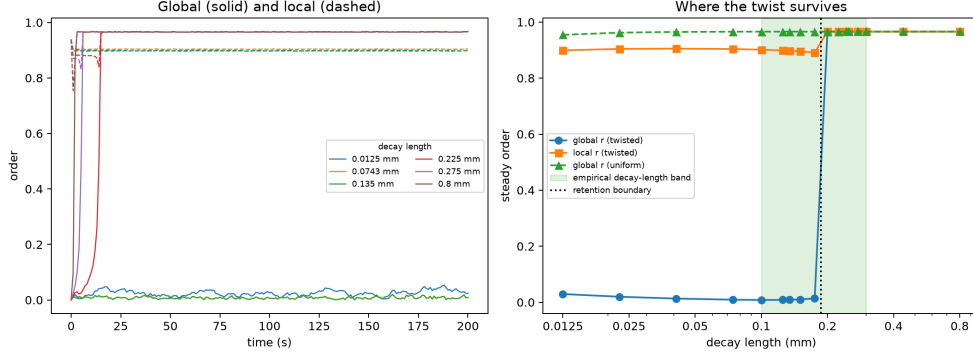


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

463 form, each separation reweighted by the cosine of the phase the winding
 464 advances across it (`hasDerivAt.charJacobian`). That reweighted kernel is
 465 again even in the separation, so the same characters diagonalize it and ev-
 466 ery mode's rate is one of its eigenvalues. Where the winding advances by
 467 at most a quarter turn across every separation the coupling reaches, the
 468 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 469 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 470 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 471 longer separations is what carries a phase advance past a quarter turn, and
 472 that is the mechanism the decay-length sweep varies. The criterion bounds

473 rates rather than trajectories: asymptotic stability is not proved, so the
 474 states that persist occupy a narrower range than it admits, and the bound-
 475 ary quoted above is measured and not derived from it.

476 The same reweighted kernel measures the potential's curvature. A wind-
 477 ing has $r = 0$ where a global minimizer of the Lyapunov potential has $r = 1$,
 478 so it is not one (`char_not_potential_min`): the quarter-turn condition on ini-
 479 tial data does not merely decline to cover a winding, it must exclude it. Along
 480 a mode, the potential's second variation is that kernel's quadratic form, so a
 481 growing mode is a direction of falling potential and a winding carrying one
 482 is not a local minimum either (`char_not_local_min_of_charLambda_pos`).

483 An amplitude layer decides which of these states exist, and leaves what
 484 the resultant reports unchanged. A phase state carries no amplitude. Ev-
 485 ery oscillator sits on the unit circle, so a field with an amplitude node has
 486 no representation in these dynamics. The real Stuart–Landau extension
 487 supplies one on the same coupling, and leaves intact the gradient struc-
 488 ture the convergence results rest on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i =$
 489 $(\mu - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its phase equation is the Kuramoto equation
 490 with each coupling weighted by a ratio of amplitudes (`amplitude_phase_is_`
 491 `kuramoto`). The phase model is this one with the amplitude ratios set to
 492 one, rather than a separate object.

493 What the added degree of freedom decides is which windings exist at all.
 494 On a periodic array of any dimension, this sheet included, a winding requires
 495 $\mu + \lambda > 0$. Here $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue the kernel supplies,

496 summed over separations, with ϕ_d the phase the winding advances across d
 497 (`exists_charState_iff`; on the ring $\phi_d = 2\pi qd/n$). For the nonnegative
 498 kernels used here $\lambda \leq 0$, and moving coupling mass to longer separations
 499 lowers it further. A broad enough kernel therefore removes the winding out-
 500 right. The sweep above reaches that same boundary by a different route:
 501 the phase model retains every winding at every coupling, and loses this one
 502 under integration instead. The two layers bound the same regime from op-
 503 posite sides. A phase recording does not separate a state the amplitude layer
 504 excludes from one the dynamics fails to sustain.

505 These states are windings and not spirals, and on the sheet that is a
 506 theorem rather than a shortage of dimensions. A cortical phase singularity
 507 is a point at which the amplitude vanishes and around which the phase turns
 508 once, read in practice as the circulation around a small plaquette. The states
 509 above have equal modulus at every site, so that circulation is exactly zero
 510 on every plaquette (`plaquetteCirculation_eq_zero_of_winding`). Nor can
 511 the layer host a single defect. Over a periodic sheet the circulations of *any*
 512 phase field sum to zero, so defects arrive in cancelling pairs (`no_isolated_`
 513 `defect`). The sweep's near-zero measured defect density agrees with the first
 514 of these and says nothing beyond it.

515 Two limits bound the layer for a reader of the cortical literature. Relax-
 516 ational dynamics has no propagation speed, which is the quantity a cortical
 517 wave measurement leads with. And $|z|$ is a normal-form radius, which a mea-
 518 sured envelope is not; this work identifies the two nowhere. Supplemental

Material gives the sweep, its uniform control, the amplitude layer and the standing wave it admits inside the band, together with the remaining limits of both.

A vanishing resultant is the absence of global agreement and not of agreement. Eq. (6) read on a nearest-neighbour patch of a winding gives a discrepancy bound of $2\sqrt{2}L|\sin\phi|$, with ϕ the phase advanced across the patch, which falls to zero as the same winding is spread over more sites (`chord_length_of_char_patch`), while the global form of the same bound is vacuous at $r = 0$. Encoders shared on overlaps therefore still bound content disagreement inside a patch of a state that no global observable calls coherent. The winding number itself stays unconnected to content.

This qualifies what a phase trace can establish. An onset criterion stated in r alone registers a change of basin as readily as a change of coupling. The protocol of Section 7.3 therefore needs a spatial observable beside the resultant.

6.4 Coherent adaptation can lose environmental alignment

An adaptive coupling rule might supply representational structure, but its objective must reward the relevant agreement. For symmetric coupling, pairwise sine terms cancel in the population mean drift. Minimizing squared drift at that fixed mean therefore rewards equalizing drift across nodes. In the

540 frequency-cluster control, this favors coupling between unlike-frequency clus-
541 ters, whereas the diagnostic environmental template rewards coupling within
542 clusters.

543 The formalization proves descent of the squared-drift objective for a spec-
544 ified gradient flow at fixed phases. Its functional differs from thermodynamic
545 entropy production: at a coherent stationary density the probability current
546 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
547 rial gives the operator, its gradient and the conditions of the descent result.

548 The finite control co-evolves noisy phases and symmetric, nonnegative
549 couplings at fixed total resource. Its template is a diagnostic covariance of
550 latent signals, never a training input. Across three seeds, second-half order
551 remains between 0.9167 and 0.9222, while the within-cluster to between-
552 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5).
553 Random-update, permuted-gradient and frozen-kernel controls are reported
554 in Supplemental Material.

555 The immediate post-update objective reduction disappears when aver-
556 aged over complete update intervals. A prespecified grid of 6 rate–cadence
557 combinations contains no case meeting the 5% sustained-reduction criterion
558 while retaining coherence and losing alignment. The finding is coherent adap-
559 tation away from the specified structure, with no demonstrated sustained cost
560 reduction.

561 Kernel alignment and overlap compatibility are different observables.
562 This control addresses acquisition of its designated template under its par-

563 ticular objective. Learning models can explicitly include data in an objective
564 (Corberi et al., 2026); an account of agreement about the cup must specify
565 the content its learning rule is meant to preserve.

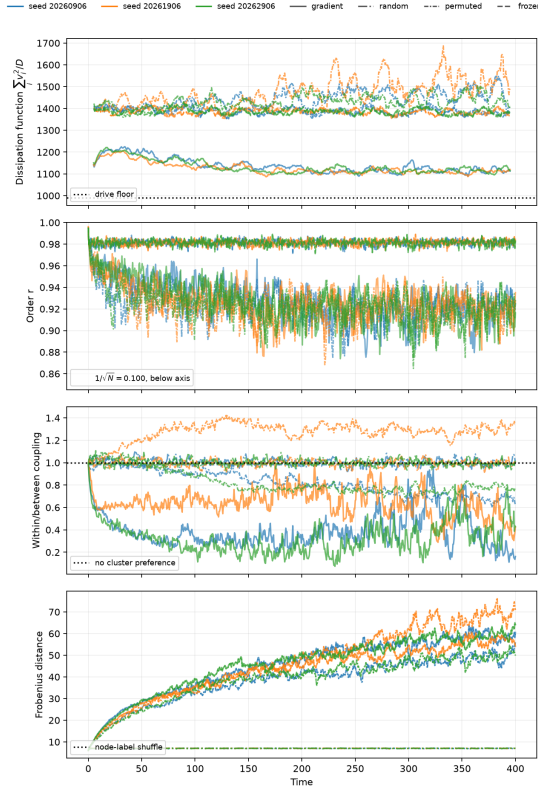


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

566 6.5 Content agreement needs independent decoders

567 A neural test needs independently identified domains, local descriptions and
568 restriction maps. Two decoders should estimate a shared content on over-
569 lapping domains before their agreement is examined. Discrepancy requires
570 a null under decoding noise and controls using deliberately compatible and
571 incompatible examples. An implementation on constructed data separates
572 these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by
573 trial; the compatible construction scores 0.476 on an interval containing one
574 half. Shared-prior shrinkage can nevertheless manufacture agreement, and
575 regional bias can manufacture disagreement. Per-patch decoding diagnostics
576 detect these failures; the compatibility statistic alone does not.

577 Phase agreement cannot replace that measurement. On constructed con-
578 tent, phase carries a share 1.000 of the content entropy when content is built
579 from it and 0.000 when it is not. A phase-derived statistic reports com-
580 patibility on phase-locked patches whose local descriptions disagree. The
581 implemented content statistic detects their incompatibility. These controls
582 validate a measurement on known constructions; a neural content model and
583 decoder remain to be established.

584 Choosing the cover also chooses a candidate subject: trivial descriptions
585 can agree while carrying little content, and a partition can conceal disagree-
586 ment. The useful choice must therefore be constrained by what neural ac-
587 tivity encodes. The exact composition assumes compatibility of its supplied
588 cover. The content mechanisms above identify additional hypotheses under

589 which agreement can arise; they leave the empirical choice of content and
590 the connection to that cover explicit.

591 **6.6 What an informative phase-density measurement** 592 **requires**

593 The concentration range and the dependence structure decide whether the
594 concentration-coherence relation can be tested at all. Under the specified
595 log-density estimator and bias-floor criterion, discrimination at the observed
596 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
597 designs, rising to 31000 when sites share a mean in groups of 10. These
598 counts describe that estimator and dependence model; the intended recording
599 requires its own calibration. Supplemental Material reports the bin-count
600 controls and the full dependence-aware design.

601 That ceiling comes from exercising the estimators on public propofol-
602 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms windows.
603 Bipolar and common-reference montages produce different residuals. The
604 bipolar results are compatible with the Bessel relation over a concentration
605 range too narrow to distinguish it from its linear approximation at the ob-
606 served scatter, which is the range the design scan takes as given. Temporal
607 pooling changes the estimand, and sedation recordings leave the awakening
608 hypothesis untested.

609 An informative dataset therefore needs sufficient concentration range or

610 precision, calibrated estimators and an observation model for dependence,
611 source mixing and reference choice. Public mouse calcium imaging provides
612 thousands of spatially located neurons across waking, sleep and anesthe-
613 sia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-
614 estimation band used here. Site count, state coverage and bandwidth are
615 separate requirements.

616 **7 The cortical hypothesis and an awakening**

617 **test**

618 The cortical proposal identifies a physical source of coupling and asks whether
619 its changes help explain recovery. The tests in Section 6 determine what
620 observations must distinguish before a phase trace can support this identifi-
621 cation.

622 **7.1 Physical identification and calibration**

623 The cortical hypothesis identifies part of effective coupling with endogenous
624 extracellular electromagnetic interactions. Fields can modulate spike timing
625 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
626 propose coordination of distributed neural information (Pinotsis & Miller,
627 2022; McFadden, 2020). Phase organization under propofol motivates study-
628 ing this regime (Bastos et al., 2021), but does not measure coupling.

629 The formal field predicate requires a jointly continuous kernel, a normal-
630 ized substrate, coupling equal to the kernel average, and no single site pro-
631 viding more than half that strength. Toy constructions satisfy it. A cortical
632 realization must specify the scale at which the kernel and its reduction ap-
633 ply: calcium imaging reports different state-dependent modularity patterns
634 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

635 The spatial reduction is also testable. A row-sum aggregation at zero
636 diagonal tracks the uniform-coupling control over the tested spatial decay
637 range, while frequency heterogeneity displaces the operational threshold. The
638 fit window biases both estimates relative to $2D$; agreement with the control
639 is therefore the supported conclusion. Supplemental Material reports the
640 reduction, its numerical error and the distinct assumptions needed for the
641 kernel-average identification in Eq. (5).

642 The field commitment requires physical evidence beyond a choice of math-
643 ematical representation. A finite matrix spread over positive-mass cells pre-
644 serves its total coupling and phase correlation exactly. Point-supported
645 vanishing under an atomless measure and a disadvantage from excluding
646 a better-correlated pair concern specified support constraints. Neither com-
647 parison supplies energy prices or a biological–digital distinction.

648 A spike-time shift alone supplies no coupling rate. The dimensionless
649 product of population size N , timing sensitivity s , field amplitude E and

650 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

651 This rate describes how induced timing shifts accumulate into drift. Nei-
652 ther it nor the installation factor κ follows from a heat bound. Testing the
653 physical threshold requires independent estimates of the interaction kernel,
654 its aggregation and diffusion in the same cortical regime.

655 **7.2 Extracellular geometry and recovery**

656 The proposed control variable is extracellular geometry. Extracellular vol-
657 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
658 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
659 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
660 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
661 an effective-medium description predicts that a larger extracellular fraction
662 raises conductivity and reduces field amplitude.

663 The coupling-gated hypothesis consequently predicts that extracellular
664 contraction on awakening raises E and, holding the other factors in Eq. (15)
665 fixed, raises effective coupling. This argument supplies a direction; the
666 geometry–coupling relation, its magnitude and its role in human recovery
667 require measurement.

668 For a smooth crossing at nonzero speed, evaluating the stationary branch

669 along the changing control gives the benchmark

$$r_{ss}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

670 This follows from Eq. (3) at fixed D and concerns the stationary branch
671 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
672 dently justified tracking approximation. The crossing controls in Section 6.2
673 leave that approximation unsupported at onset, even though a separation of
674 geometry and phase time scales can support it away from threshold. The
675 awakening hypothesis therefore needs a calibrated dynamical prediction for
676 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
677 does not supply a recovery latency.

678 **7.3 A protocol for the awakening experiment**

679 The physical test has six requirements:

- 680 **1. Coupling and diffusion in the same units.** Estimate them from re-
681 solved phase dynamics, separating heterogeneity and observation noise.
682 Measure the drive or perturb the system to distinguish endogenous cou-
683 pling.
- 684 **2. Geometry calibrated to coupling.** Validate the geometry observ-
685 able and its relation to effective coupling, with uncertainty. The con-
686 ductivity argument supplies only the expected direction.

- 687 **3. Calibrated phase measurements.** Estimate r by resultant length
688 and a by unweighted regression of log density on centered cosine.
689 Choose site count and binning from a dependence-aware calibration
690 over the intended concentration range. The tested designs require
691 thousands of sites at the observed concentrations; temporal pooling
692 changes the spatial estimand.
- 693 **4. An independent recovery endpoint.** Measure behavioral recovery
694 separately from phase organization.
- 695 **5. A predeclared dynamical onset criterion.** Set a coherence thresh-
696 old above the finite-size floor and evaluate its latency under the mea-
697 sured coupling trajectory, including ramp speed or fluctuations. Record
698 a spatial observable beside the resultant, since a winding field reports
699 a low resultant at coupling that sustains local order (Section 6.3).
- 700 **6. Matched model comparison.** Compare coupling increase, diffusion
701 change, common drive and alternative recovery models under the same
702 observation and filtering assumptions.

703 Failure of the concentration–coherence relation over an informative range
704 constrains the density model. Failure of the calibrated geometry–coupling
705 association constrains the physical gate. Better prediction of independent
706 recovery by a competing mechanism counts against its proposed role. A
707 favorable outcome still leaves the content of coordinated activity to be ex-
708 plained.

709 8 Discussion

710 The account set out in Section 1 is conditional at every step: each connection
711 in it is a hypothesis supplied rather than a result derived, and Table 1 names
712 all eight. What follows takes those conditions in turn, as commitments the
713 proposal incurs, as measurements that would bear on them, and as questions
714 the framework leaves open.

715 8.1 Unity and a minimal self as empirical commit- 716 ments

717 We propose the state glued from compatible local descriptions as a corre-
718 late of conscious unity, and its accurate internal reconstruction as a correlate
719 of a minimal self. The motivations are that different contents belong to
720 one experienced situation and that the situation is experienced from a per-
721 spective. The conditional results identify relations a realization would need:
722 sufficient installed resources for the specified phase model, agreement about
723 shared content, and an encoding that supports reconstruction at the required
724 fidelity.

725 Compatibility allows different descriptions to contribute different infor-
726 mation. It also permits uncertainty and conflicting judgments: apparent cup
727 position and judged cup position can be distinct quantities in a consistent
728 description of their discrepancy. Which quantities are shared must be de-
729 termined independently. Choosing them solely to secure agreement would

730 deprive the proposal of empirical force.

731 The minimal-self proposal concerns the perceiver’s position and condition
732 within the situation, such as the cup’s location relative to a body that can
733 reach it. The encoding could be implicit and need carry no autobiograph-
734 ical narrative. Its proposed necessity is philosophical: even an unreflective
735 episode has a perspective. Gluing by itself requires no self-encoding, and re-
736 constructability alone supplies no functional role. Altering the encoded hand
737 or viewpoint while holding the external scene fixed should change estimates
738 of the perceiver’s relation to the cup as the readout predicts. Whether that
739 represented perspective is experienced remains a further question.

740 **8.2 What a test of the full proposal must establish**

741 Both uniqueness results concern specified objects. They choose neither the
742 cover nor the encoding, and do not select between candidate subjects. In-
743 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
744 here selection requires independent evidence about neural organization. A
745 constant readout can contract and fix a predetermined state while discard-
746 ing its input. The family criterion in Section 4.2 tests the sensitivity that
747 this construction lacks, provided the relevant states and fidelity are declared
748 independently.

749 A test must separately measure coupling relative to diffusion, content
750 compatibility and self-reconstruction, alongside an independent endpoint for
751 experience. Meeting all three without a unified episode would challenge suffi-

752 ciency. A unified episode with incompatible descriptions would challenge the
753 unity identification; one lacking the specified self-encoding would challenge
754 the reflexive requirement. Awakening, dreaming and dissociative states offer
755 possible settings, but absence of report cannot by itself define absence of
756 experience.

757 The next physical requirement is a calibrated geometry–coupling relation
758 and resolved phase dynamics. The installation bound contributes a nec-
759 essary resource condition where its hardware representation applies; cortical
760 mode prices and field-mediated coupling require measurement. For the repre-
761 sentational connection, an informative experiment asks whether a controlled
762 change in coordination alters independently decoded content agreement, with
763 controls for sensory information and decoder quality. The constructed-data
764 estimator establishes feasibility for specified contents. Neural local descrip-
765 tions, shared overlap encoders and the self-encoding have yet to be identified.
766 The population-size dependence of the phase-to-content bound further limits
767 its quantitative use.

768 The separation results explain why these measurements must be inde-
769 pendent. A stationary distribution leaves several recovery mechanisms indis-
770 tinguishable. Equal phase order can accompany incompatible descriptions
771 or different task information. The tested plasticity rule preserves coherence
772 while losing its designated template. Each finding concerns the specified
773 model and observable; their role is to delimit the evidence needed for the
774 proposed connections.

775 8.3 GPUs and the limits of no-go claims

776 For a large language model (LLM) running on graphics processing units
777 (GPUs), the framework requires identifying a physical system whose local
778 descriptions form one state and whose internal encoding reconstructs that
779 same state. The coupling-gated route E56–E78 requires a calibrated physical
780 coupling, its noise regime and a connection from coordination to content
781 agreement. An attention matrix or a shared clock supplies none of these identifications on its own. Computing a field model likewise leaves its realization
782 in the computing substrate to be established. These physical obligations also
783 apply to the cortical proposal.

785 Unity requires the candidate’s local descriptions to agree on independently identified shared quantities (Section 3). Common timing and fluent
786 output do not establish that agreement. Digital representations can satisfy compatibility, however, so finite hardware is compatible with the gluing
787 construction. The empirical commitment is to identify the relevant cover and contents with one experienced situation; mathematical uniqueness alone
788 selects no subject.

792 For self, the system must carry an encoding $E(s)$ and an internal readout
793 R of the same global state. E89 requires $F = R \circ E$ to contract and fix
794 the state produced by gluing. A constant readout shows why a unique fixed
795 point alone is insufficient for informative self-representation. The additional
796 family criterion in Section 4.2 supplies a test: if two relevant macrostates
797 satisfy $E(s) = E(t)$ and $d(s, t) > 2\varepsilon$, Eq. (10) rules out reconstructing both

798 within error ε . An intervention outside the encoding region’s causal past
799 gives a corresponding obstruction before a declared communication deadline.
800 Identical verbal reports alone do not establish identical internal codes. These
801 conditional no-go results exclude a specified reconstruction at a specified
802 fidelity or deadline. They apply to biological and digital candidates alike.

803 The hardware comparisons in Supplemental Material have similarly re-
804 stricted scope. Fixed wiring can impose a correlation disadvantage, and
805 point-supported kernels vanish under an atomless continuum measure. Nei-
806 ther comparison makes the representation conditions unattainable on GPUs.
807 The installed-energy condition of Section 2.2 is restricted in the same way.
808 Once a candidate’s installed energy stands far above the thermal floor, which
809 it does for any macroscopic substrate, the condition constrains that candidate
810 only through κ , the conversion from installed energy to coupling; κ is hard-
811 ware data supplied by mode profiles and prices, and no result here derives
812 it for cortex or for a processor. Supplemental Material reads the condition
813 against a cortical metabolic budget and returns the conversion factor it would
814 require rather than a verdict. A universal exclusion would need a necessary
815 condition for consciousness and a proof that every relevant implementation
816 violates it. The present conditional composition supplies neither.

817 Whether experience requires a living substrate is the explicit subject of
818 biological naturalism (Seth, 2026a,b). On that view the properties ground-
819 ing experience are those of self-sustaining organisms, and simulating them
820 does not secure them. The conditions stated here name no biological mecha-

nism. Computation described as such does not satisfy them either: a candidate must exhibit a physically realized coupling in a measured noise regime, agreement about independently identified shared quantities, and an internal reconstruction meeting a declared fidelity. Whether a realization of those conditions that is not alive would be accompanied by experience is the point at issue, and these results do not settle it. A related position locates the difference in the mode of computation rather than in life as such: biological systems compute across scales that do not separate, and alongside discrete operations carry continuous-valued dynamics belonging to their physical medium (Milinkovic & Aru, 2026). The coupling condition stated here is one measurable instance of that requirement, fixed to a physical field in a declared noise regime, and a description of such a field does not satisfy it.

The neuroscientific case against *human-like* consciousness in text-only LLMs concerns embodied sensory interaction, thalamocortical organization and biological regulation linking perception and action to survival (Aru et al., 2023). Their absence supports skepticism, conditional on their causal relevance to experience. Those mechanisms are not requirements of the abstract gluing and reconstruction definitions. Workspace-like representations reported in some LLMs (Gurnee et al., 2026) provide candidate organization to investigate. Within this framework, the relevant evidence would identify compatible contents and an informative internal reconstruction, or demonstrate their failure under controlled interventions.

8.4 Explanatory scope

Integrated-information, global-workspace and predictive-processing accounts organize unity, access and modelling differently (Tononi et al., 2016; Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a candidate physical interaction and conditional relations between local content and self-encoding. Gluing and reconstruction permit multiple substrates; the electromagnetic commitment enters through physical calibration. The formalization checks implications under their hypotheses, and its witnesses establish realizability on declared models. They do not together supply a cortical agent.

The explanatory question is whether the specified relations occur in neural activity and matter for experience. In the cup example, this means establishing how coordination helps maintain one represented situation and how the perceiver’s place within it contributes to the episode. The proposal identifies candidate correlates and tests of that identification. Why those relations would be accompanied by experience remains unexplained.

Data and code availability

The Lean development, simulation scripts, tests and saved numerical summaries are maintained in the GitHub repository `bobaseb/coupling_gated_cortical_coherence`. Publication simulation values are generated from saved summaries without rerunning production sweeps. The exploratory

864 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
865 processing and selection are described in Supplemental Material. The repos-
866 itory contains the reproducible analysis and does not redistribute the raw
867 recordings.

868 **Funding and competing interests**

869 This research received no external funding. The author declares no compet-
870 ing interests.

871 **Use of AI tools**

872 AI assistance to this work was substantial and is not separable by task. Sev-
873 eral coding assistants and chat interfaces were used in an interleaved way
874 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
875 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
876 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
877 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
878 accessed with the Hermes CLI. Between them they contributed to the Lean
879 development, the simulation and analysis code, and the drafting and revision
880 of both documents. Because the tools were used in combination and in alter-
881 nation, no section, proof, figure or result is attributable to any one of them,
882 and none is claimed to be. The author is responsible for all content, including

883 everything a tool produced. Machine checking establishes the stated Lean
884 results under their hypotheses; it does not validate biological interpretations
885 or replace scientific review of generated code and prose.

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